

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

.....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

.....
COMMISSION NATIONALE D'ORGANISATION DE
L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

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MINISTRY OF HIGHER EDUCATION

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NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

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National Exam of Higher National Diploma 2024 Session

Speciality/Option: Software Engineering (SWE)

Paper : Case Study

Duration : 6 hours

Credit : 14

ANSWER ALL QUESTIONS

SECTION A: ALGORITHM AND PROGRAMMING (50 MARKS)

Question 1 Algorithms (10 Marks)

// This program uses an array to display five names.

Constant Integer SIZE = 5

Declare String names[SIZE] = "Meg", "Janai", "Sonat", "Belle", "Molak"

Declare Integer index

For index = 0 To SIZE

Display names[index]

End For

Work required:

- a) Find the bug in the above-mentioned segment of code **5 marks**
- b) Implement in the language you master properly **5 marks**

Question 2 Pseudocode (25 Marks)

Consider the following pseudocode

// This program displays the highest value in the array.

Declare Integer SIZE = 3

Declare Integer values[SIZE] = 1, 3, 4

Declare Integer index

Declare Integer highest

For index = 0 To SIZE - 1

If values[index] > highest Then

Set highest = values[index]

End If

End For

Display "The highest number is ", highest

Work required:

- a) Explain the bug in the above-mentioned segment of code and fix it **10 marks**
- b) Implement the previous pseudocode in the language you master most. **15 marks**

Question 3 Object oriented programming (15 marks)

You are asked to design a program for Wireless Solutions, a business that sells cell phones and wireless service. The program will be used to keep track of the cell phones that the company has in inventory. The data that we need to keep for a cell phone is as follows:

- The name of the phone's manufacturer
- The phone's model number
- The phone's retail price

You are designing an object-oriented program, so we will create a class that represents a cell phone. The class will have fields to hold these items of data.

Work required:

- a) Declare the data fields of the class cell Phone. **3 marks**
- b) Declare the mutators (setters) and the accessors (getters) of the class cell phone. **3 marks**
- c) Declare a default constructor and a constructor with three parameter lists of the class cell phone. **3 marks**
- d) Define the class cell phone in the language you master better. **3 marks**
- e) Implement a test program for the class cell phone. **3 marks**

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (20 MARKS)

Consider the relations relational schema of a database, where the primary keys are underlined and foreign keys with (*).

User

<u>Id</u>	<u>Name</u>	<u>Age</u>	<u>Gender</u>	<u>OccupationId*</u>	<u>CityId*</u>
<u>1</u>	<u>John</u>	<u>25</u>	<u>Male</u>	<u>1</u>	<u>3</u>
<u>2</u>	<u>Sara</u>	<u>20</u>	<u>Female</u>	<u>3</u>	<u>4</u>

Occupation

<u>OccupationId</u>	<u>OccupationName</u>
1	Software
2	Accountant
3	Pharmacist

City	
CityId	CityName
1	Bertoua
2	Buea
3	Bamenda

- Write relational algebra and SQL statement that outputs the names of the users who are more than 25 years old. **5 marks**
- Write relational algebra and SQL statement that outputs all information of users who have an Id lesser than 3 or who are of Age different from 32. **5 marks**
- Write algebra and SQL statement that outputs all information of users together with information about their occupation. **5 marks**
- Write relational algebra and SQL statement of user's names and gender who work in Bamenda. **5 marks**

SECTION C: WEB DESIGN 15 MARKS

Learn to position HTML elements using CSS positioning properties.

Code:

```
<!DOCTYPE html> <html>
<head>
<style>
.relative { position: relative; left: 20px;
top: 10px;
}
.absolute {
position: absolute;
top: 50px;
right: 30px;
}
.fixed {
position: fixed;
bottom: 0;
right: 0;
}
</style>
</head>
<body>
<div class="relative">Relative Positioning</div>
<div class="absolute">Absolute Positioning</div>
```



```
<div class="fixed">Fixed Positioning</div>
</body>
</html>
```

Work required:

- | | |
|--|----------------|
| a) Name the three positioning elements covered in the above code | 3 marks |
| b) Explain the function of each positioning element | 3 marks |
| c) Provide a title for this code | 3 marks |
| d) What is the purpose of this code? | 3 marks |
| e) Define the function of CSS in the HTML file | 3 marks |

SECTION D: NETWORKING (15 MARKS)

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|---|----------------------|
| 1) Explain the importance of layering in a network | 2 marks |
| 2) Differentiate between each of the following | |
| a) OSI model and TCP/IP model | 2 marks |
| b) UDP and TCP | 3 marks |
| 3) Describe detailly the data encapsulation process in OSI | 2 marks |
| 4) For each of the following network give a sketch, two advantages ,two disadvantages and a network protocol or topology employing it | |
| a) Mesh | |
| b) star topology | 2×3 = 6 marks |